NR2DT-01: Step 1 — Discover

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 1 of 10 | **Created:** April 2026 | **Last Updated:** 08/27/2026

Overview

Goal of this step: produce a complete inventory of the source New Relic account, a translation-confidence report, and a defensible effort estimate. Nothing else moves until discovery is complete.

This is procedural. For the *why* behind each artifact and the deeper translation theory, see **NRLC-02** (NRQL→DQL Translation) and the relevant NRLC component notebook for each entity class.

Sprint 1.337 (April 2026) Updates Affecting NR2DT

Sprint 1.337 introduces three changes worth surfacing during the discovery step:

1. **OneAgent primary fields/tags at source** — when designing the post-migration tag taxonomy (see NR2DT-03 Design), prefer OneAgent primary tags over OpenPipeline parsing where possible. Hosts can emit `team`, `cost_center`, `env` as top-level fields via `oneagentctl --set-host-tag`.
2. **OTel `service.name` enrichment** — if the New Relic source uses OTel auto-instrumentation, services map cleanly to Dynatrace without the historical two-services-per-process pattern (NRLC-01 § Sprint 1.337 covers the deeper detail).
3. **Settings v2 + Platform tokens** for new automation (NR2DT-04 Translate, NR2DT-09 Cutover scripts).

Engine impact: the Dynatrace-NewRelic conversion engine and lab-setup scripts in `topics/nr2dt/lab-setup/` were validated end-to-end on a live Gen3 sprint tenant 2026-04-20 — re-verify before each new engagement that the conversion engine handles sprint-337's API removals (Entity API `attributes`, Event API `metadata`).

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-01 (Platform Comparison) for orientation

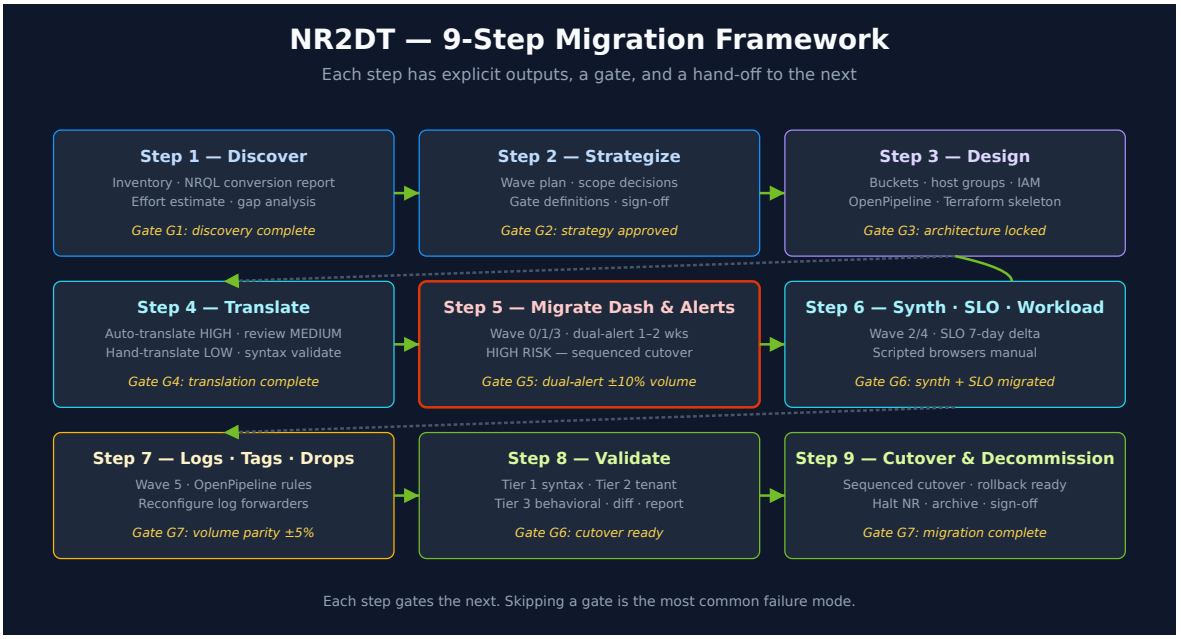
1. What You'll Produce

By the end of this step you will have these artifacts checked into your migration repo:

Artifact	Format	Purpose
`inventory/exports/newrelic_export.json`	JSON	Full enumeration of all NR entity classes
`nrql-conversion-report.csv`	CSV	Every NRQL query with HIGH/MEDIUM/LOW confidence + DQL output
`effort-estimate.md`	Markdown	Hours/days/weeks per component class

Artifact	Format	Purpose
`gap-analysis.md`	Markdown	Entities flagged for manual work (APM conditions, scripted browsers)
`risk-register.md`	Markdown	Known risks with proposed mitigations

These artifacts gate Step 2 (Strategize). Don't proceed without them.



2. Prerequisites & Access

NR API token

User-level API key with **read** scope on:
- Dashboards, Alerts, NRQL, Synthetics, SLO, Workloads, Logs

```
export NR_API_KEY="NRAK-..."
export NR_ACCOUNT_ID="&lt;numeric&gt;"
export NR_REGION="US"  # or EU
```

DT platform token

Step 5 (effort report) and Step 8 (validation) connect to the target Dynatrace tenant, so you also need DT credentials configured before you run them:

```
export DYNATRACE_API_TOKEN="dt0c01.XXXXX..."  # platform token recommended
export DYNATRACE_ENVIRONMENT_URL="https://&lt;env-id&gt;.live.dynatrace.com"
```

Store these in `.env` rather than the shell for long-running work. Verify with:

```
python3 migrate.py preflight
```

`preflight` probes the tenant for Settings 2.0 / Document API / Automation API availability. Green here means Step 5+ will actually connect.

Tooling

Clone the migration framework:

```
git clone https://github.com/timstewart-dynatrace/NewRelic-to-Dynatrace-
Migration-Utilities
cd Dynatrace-NewRelic
pip install -r requirements.txt
cp .env.example .env  # populate NR_* vars
```

3. Run the Inventory

Command:

```
python3 migrate.py migrate --export-only --output ./inventory
```

This runs the NerdGraph enumeration for every entity class and writes per-component JSON files plus the consolidated `inventory/exports/newrelic_export.json`.

Check the output:

```
ls -lh inventory/
cat inventory/exports/newrelic_export.json | jq '. | keys,
[.dashboards|length, .alert_policies|length, .synthetic_monitors|length,
.slos|length, .workloads|length]'
```

Expected counts to capture (your numbers, for the wave plan in Step 2):

- Dashboards (and total widget count across all pages)
- Alert policies + NRQL conditions + APM conditions
- Synthetic monitors by type (Ping / Browser / API / Step)
- SLOs
- Workloads
- Notification channels by type
- Drop rules + log parsing rules + tag rules

Extract NRQL for the translation pass

`compile` and `batch` (Step 4) need a flat list of NRQL queries. The export JSON has the NRQL embedded inside dashboards, alert conditions, and SLO indicators — use the `extract-nrql` subcommand to pull them into a file:

```
# Plain text (one NRQL per line — feeds `compile --file`):
python3 migrate.py extract-nrql \
    --input ./inventory \
    --output ./inventory/all-nrql.txt

# Or CSV with an `nrql` column (feeds `batch --file`):
python3 migrate.py extract-nrql \
    --input ./inventory \
    --output ./inventory/all-nrql.csv
```

The subcommand walks dashboards, alert conditions, and SLO indicators; dedupes; writes either format depending on the output extension (override with `--format txt|csv`). Capture the query count from the success line — that's the denominator for Step 4's confidence distribution.

▶ **Fallback for older tool versions** (without `extract-nrql`)

4. Translate the NRQL Surface

Every dashboard widget, alert condition, and SLO indicator contains a NRQL query that must compile to DQL. Run the compiler now to size the translation effort.

Command (uses the `inventory/all-nrql.txt` or `inventory/all-nrql.csv` produced by `extract-nrql` in §3):

```
# (a) compile — writes translated DQL, one per input query
python3 migrate.py compile --file inventory/all-nrql.txt --output nrql-translated.dql

# (b) batch — writes a CSV with nrql / dql / confidence / warnings columns.
#     Feed the .csv you created in §3 directly (or convert a .txt with awk):
python3 migrate.py batch --file inventory/all-nrql.csv --output nrql-conversion-report.csv
```

(If you extracted to `.txt` in §3 instead of `.csv`, convert with: `awk 'BEGIN{print "nrql"} {print "\"" $0 "\""}' inventory/all-nrql.txt > inventory/all-nrql.csv`)

Which artifact matters for the G1 gate? The CSV from `batch` — it has the confidence distribution you need to compare against the table below.

Output:

- `compile --output <file>` writes a plain DQL file (one translated query per input).
- `batch --output <file.csv>` writes a CSV with one row per query: source NRQL, translated DQL, confidence (HIGH/MEDIUM/LOW), confidence score (0–100), notes, warnings. **This is the artifact to review for the Step 1 gate.**

Expected confidence distribution for a typical NR account:

Confidence	Typical share	Action in Step 4
HIGH	70–80%	Auto-translate; sample-validate during dual-run
MEDIUM	15–25%	100% review during dual-run
LOW	0–10%	Hand-translate or reformulate

If LOW is > 15%, surface this to stakeholders early — it's the dominant cost driver.

Deep-dive reference: NRLC-02 explains the compiler architecture, the 292 patterns, and how confidence is scored.

5. Generate the Effort Estimate

5a. Run the report generator (connects to DT)

```
python3 migrate.py migrate --report --input inventory --output effort-
estimate.md
```

> **Requires DT credentials** — `DYNATRACE_API_TOKEN` and `DYNATRACE_ENVIRONMENT_URL` from §2. If they aren't configured yet, the command exits with "Failed to connect to Dynatrace" and doesn't write the file.

5b. Offline fallback (no DT connection required)

If your Step 1 run happens before DT access is provisioned — typical for the first discovery sprint — derive the estimate offline from the inventory + conversion CSV:

```
python3 - &lt;&lt; 'PY' &gt; effort-estimate.md
import json, csv, pathlib

inv =
json.loads(pathlib.Path('inventory/exports/newrelic_export.json').read_text())
rows = list(csv.DictReader(open('nrql-conversion-report.csv')))

print('# NR→DT Migration — Discovery Artifact (Step 1)\n')
print('## Inventory counts\n')
for k in ('dashboards', 'alert_policies', 'synthetic_monitors', 'slos',
'workloads'):
    print(f'- **{k}**: {len(inv.get(k, []))}')
print('\n## Translation confidence distribution\n')
conf = {}
for r in rows: conf[r["confidence"]] = conf.get(r["confidence"], 0) + 1
total = sum(conf.values())
for k, v in sorted(conf.items()):
    pct = 100 * v / total if total else 0
    print(f'- **{k}**: {v} ({pct:.0f}%)')
print(f'\n## Totals\n\n- NRQL queries: {total}')
PY

cat effort-estimate.md
```

Both paths (5a and 5b) produce an `effort-estimate.md` that satisfies the G1 exit criterion. Use 5a when DT access is ready; use 5b during early discovery.

Estimating heuristic per component class:

Class	Light (hrs)	Moderate (days)	Heavy (weeks)
Dashboards	< 20	20–80	80+
Alert conditions	< 50	50–200	200+
Synthetics (incl. scripted)	< 10	10–40	40+
SLOs	< 5	5–20	20+
Notification channels (with secrets)	< 5	5–20	20+

6. Step Exit Criteria

G1 — Discovery Complete

Pass criteria — all five must be true:

- [ ] `inventory/exports/newrelic_export.json` complete; every entity class has a non-empty array; no enumeration errors in log
- [ ] Per-component counts captured and documented
- [ ] `nrql-conversion-report.csv` generated; LOW count <= 15% of total
- [ ] `effort-estimate.md` reviewed by lead
- [ ] `gap-analysis.md` reviewed; manual-work items have estimated owners

If any check fails, do not proceed to Step 2. Fix the gap first.

Next step: NR2DT-02 — Strategize (wave planning, scope decisions, gate definitions for the rest of the migration).
